

# Muhammad Saran

## Senior AI Software Engineer

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### Profile

I'm a Full Stack & AI Engineer with 3+ years of experience designing, developing, and deploying scalable software solutions powered by modern web technologies and Artificial Intelligence. My expertise spans AI/ML systems, Generative AI applications, web platforms, and enterprise-grade solutions, enabling businesses to leverage data-driven intelligence and automation.

I specialize in building end-to-end applications using Python, Django, React, Next.js, and modern cloud-native architectures. In the AI domain, I have hands-on experience developing Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), and Generative AI solutions using frameworks and libraries such as TensorFlow, PyTorch, Scikit-learn, OpenCV, Hugging Face Transformers, LangChain, Pandas, NumPy, and Large Language Models (LLMs).

My work includes developing intelligent systems for image classification, object detection, plant disease diagnosis, recommendation engines, AI-powered automation, and Generative AI applications. I am passionate about combining robust software engineering practices with cutting-edge AI technologies to build secure, scalable, and high-performance solutions that deliver measurable business impact.

### Skills

Python, JavaScript, SQL, Django, Django REST Framework (DRF), Flask, FastAPI, HTML5, CSS3, Bootstrap, Tailwind CSS, PostgreSQL, MySQL, MongoDB, Machine Learning, Deep Learning, Generative AI, Large Language Models (LLMs), Computer Vision, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, Hugging Face Transformers, TensorFlow, PyTorch, Scikit-learn, OpenCV, Pandas, NumPy, Vector Databases, REST APIs, AWS, Git, Jira.

### Education

**COMSATS University**, *Bachelor's Degree in Computer Science* 06/2018 – 06/2022 | ABT, Pakistan

- ♦ Focus on Software Development, Algorithms, and Distributed Systems
- ♦ Final Year Project: DeepScan AI – Cross-platform in web, mobile, and AI in Media Forensics Platform using React, Python, Django, TensorFlow/PyTorch, Flutter.

### Experience

**Exarth**, *Senior AI Software Engineer*

01/2025 – 05/2026

Islamabad, Pakistan

- ♦ Led multiple AI-driven projects, including RAG systems, chatbot assistants with LangChain and Rasa, and LLM integrations with OpenAI (ChatGPT), Claude, and Gemini APIs.
- ♦ Built a RAG pipeline using LangChain + Pinecone for a legal firm's document assistant. Achieved 80% improvement in document query response accuracy.
- ♦ Deployed multilingual customer service chatbot using OpenAI and Claude, integrated with Rasa. Supported over 6 languages and reduced human intervention by 70%.
- ♦ Led the implementation of multi-modal AI features using DALL-E and Hugging Face transformers, reducing image generation processing time
- ♦ Architected and deployed scalable applications using Python (FastAPI, Flask), Node.js, and Nest.js on cloud platforms (AWS, Azure).
- ♦ Developed pipelines using Pinecone and other vector databases to implement semantic search and knowledge retrieval systems.
- ♦ Built data preprocessing flows using NLP libraries from Hugging Face and integrated with transformer models like Llama.

- Developed scalable AI-powered web applications using Python, Django, Django REST Framework (DRF), and FastAPI, building secure RESTful APIs and enterprise-grade backend systems.
- Designed, trained, and deployed Machine Learning and Deep Learning models for computer vision, image classification, object detection, and predictive analytics using TensorFlow, PyTorch, Scikit-learn, and OpenCV.
- Built Generative AI solutions leveraging Large Language Models (LLMs), LangChain, Vector Databases, and Retrieval-Augmented Generation (RAG) to create intelligent chatbots, AI assistants, and automated knowledge systems.
- Developed responsive web applications using HTML, CSS, Bootstrap, Tailwind CSS, and JavaScript while integrating AI models into production-ready platforms to deliver intelligent user experiences.
- Deployed and maintained cloud-based applications using AWS, Docker, Git, Linux, and CI/CD pipelines, collaborating with cross-functional teams in Agile/SCRUM environments to deliver scalable and high-performance solutions.

## Projects

### DeepScan AI – Media Forensics Platform

AI-Powered Media Verification: Developed a web and mobile platform with Authentication APIs to handle secure media uploads and sync data from mobile to web. Implemented AI models to detect manipulated images and videos by analyzing faces, textures, motion artifacts, and noise patterns, providing real-time authenticity predictions with confidence scores. Built using Python, Django, TensorFlow/PyTorch

### NetSuite-ERP

Developed a comprehensive SaaS-based web ERP dashboard focused on financial management and accounting workflows. Designed and implemented core features, including the chart of accounts, account tracking, ledger management, and transaction monitoring. Built interactive charts, graphs, and reports to deliver real-time financial insights. Enabled users to analyze trends, track expenses, and make informed, data-driven decisions through a scalable, multi-tenant platform

### Caruma-TriadPay

A modern fintech application designed to enable secure and efficient digital transactions. I developed the frontend using React, building responsive and dynamic user interfaces for a seamless user experience. On the backend, I utilized Node.js to create scalable APIs and handle core business logic, including authentication and transaction processing. I integrated Convex to manage real-time data synchronization, serverless functions, and database operations. I also ensured data integrity and system security while optimizing performance for high reliability.

### Voyage Hub – Travel Booking & Services Platform

Implemented a full-featured marketplace platform for travel services with Stripe payment integration and SMTP email notifications. Implemented dynamic features using Django, Whisper mixins, and AJAX for real-time interactions, bookings, and secure transactions, delivering a seamless user experience

### StreamChat – Real-Time Chat Application

Built a full-featured real-time chat application using Django and Django Channels, enabling secure private and group messaging with live online user tracking. Implemented WebSocket-based communication for instant message delivery, user presence management, and scalable real-time interactions, ensuring high performance and reliability